Summer Internship Project Report

" Real Time Object Tracking System Using YOLOv8 And DeepSORT Alogirthm"

Report submitted in partial fulfillment of the requirements

for the award of the degree of

Bachelor of Technology in
Computer Science and Engineering (AI – ML)

Under the Guidance of
Dr.Anubhava Shrivasta
Assistance Professor, CSE Dept, FEST

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Department of Computer Science and Engineering (AI – ML)
July 2026

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

CERTIFICATE

OF INTERNSHIP

This certificate is awarded to

Dhruv Hiteshbhai Trivedi

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 2 Month AICTE & ICAC Approved internship program as

Artificial Intelligence Intern

Conducted from 01/06/2026 to 01/08/2026

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Dr. Anurag Shrivastava

Program Manager

Student’s Declaration

I Dhruv Trivedi a student of B. Tech. in the discipline of CSE (AI-ML) at Adani University, hereby solemnly declare that the report entitled:

“Real-Time Object Tracking System Using YOLOv8 And DeepSORT Algorithm”

submitted by me in partial fulfillment of the requirements for the degree of B. Tech., is my original work and has not been submitted previously, in part or full, to any other university or institution for the award of any degree or diploma.

I further affirm that the work embodied in this report does not violate any existing copyright or intellectual property rights. Wherever necessary, I have acknowledged and properly cited the work of others.

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1AUA23BCS042

Date:04/08/2026

Faculty / Mentor Certificate

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This is to Certify that this Summer Internship Project Report titled " Real-Time Object Tracking System Using YOLOv8 And DeepSORT " Algorithm" is the bonafide work of <Name of Student (Enrolment No.)>, who has carried out his / her project under my supervision. I also certify that to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate. I have also checked the plagiarism extent of this report which is ……… % and it is below the prescribed limit of 25%. The separate plagiarism report in the form of html /pdf file is enclosed with this.

Signature of the Faculty Guide/s

DR.Anubhava Srivastava,AssociateProfessor,

CSE Department, FEST

Signature of Head of Department

Asst. Prof. (Dr.) Vaishali Chourey,
Assistant Professor & Head
CSE Department, FEST

List of Tables

List of Symbols, Abbreviations or Nomenclature

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
YOLO	You Only Look Once
SORT	Simple Online and Realtime Tracking
DeepSORT	Simple Online and Realtime Tracking with a Deep Association Metric
Re-ID	Re-Identification (Deep Appearance Vector)
NMS	Non-Maximum Suppression
IoU	Intersection over Union
mAP	Mean Average Precision
FPS	Frames Per Second
OpenCV	Open Source Computer Vision Library
CPU / GPU	Central Processing Unit / Graphics Processing Unit

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Acknowledgements

First of all, I wish to thank Codec Technologies, Maharashtra, for giving me the amazing opportunity of doing mySummer Internship Program as an AI Intern from Jun 1, 2026 to Aug 1, 2026.

I wish to express my gratitude to Dr.Anurag Shrivastava and my mentor Codec Technologies for allthe guidance and criticism that I have received from them throughout the internship period. This mentoringprocess was instrumental in improving my practical knowledge in the field of deploying deep learningmodels and computer vision pipelines at the production stage.

Finally, I wish to thank my family and friends for their constant motivation and support throughout myengineering career.

Dhruv Trivedi

Table of Contents

Abstract

This document is a report of the work done during the Summer Internship Program conducted by Codec Technologies,Maharashtra from 01 June 2026 till 01 August 2026 as an AI Intern. In this duration, experience was gained in the field of machine learning algorithms, deep learning neural networks, AI chatbot, Flask web UI, demand forecasting, and computer vision.

The main project done during the internship period was the design and development of an automated fast Real Time Object Tracking System Using YOLOv8 & DeepSORT Algorithm. Monitoring of live video feeds through human effort is both time-consuming and error-prone. To overcome this drawback, a complete computer vision framework was created incorporating the use of Ultralytics YOLOv8 for object detection in a single pass and DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric) for estimating the temporal state and identity tracking of the object across consecutive video frames.

The main implementation is based on Python 3.10, OpenCV to obtain video streams in real-time and draw bounding boxes, PyTorch as a neural networks engine, and 128D Deep ReID Convolutional Neural Networks for appearance recognition purposes. Motion and spatial relationships between the objects are estimated by a combination of the Kalman Filter with the Hungarian Algorithm and cascade matching technique.

Using the above software-level optimizations, such as choosing YOLOv8n (nano version), frame skipping, and tweaking the parameters of tracking (for example, setting the maximum track age to max_age=30, Intersection over Union to IoU=0.45), we have got 89% mAP in detection, 95% Track ID consistency, and real-time performance with 25-30 FPS on CPU and under 40 ms latency per frame.

Chapter 1: Introduction

Computer vision and Automated Video Surveillance have become important technological foundations within a number of industries, such as Intelligent Transportation Systems, Smart Retail Analytics, Defense Security, and Autonomous Robotics. Today's organizations need real-time automated video analysis capabilities in order to analyze the constant stream of video feeds, recognize key objects, analyze traffic dynamics, and gain situational awareness.

Traditional object detection algorithms analyze video frames one by one. Yet, the video analysis process involves the notion of tracking that implies linking spatial objects between consecutive video frames in a way that preserves unique Track ID numbers. In order to track objects continuously in real time, an intelligent system should incorporate fast object detection algorithms as well as efficient multi-object tracking algorithms.

1.1 Objectives of the Internship

• AI/ML Engineering Practicum: For developing practical experience in implementation, training and deployment of deep learning models in an engineering environment within a company.
• Development of Computer Vision Pipeline: For constructing a computer vision pipeline which can process live webcam streams or MP4 videos with high Frame Rate (≥ 25FPS).
• Implementation of Object Detection: For implementing Ultralytics YOLOv8 (YOLOv8n, more precisely) in order to accomplish real-time single-shot object detection and classification.
• Multi-Object Tracking and Re-ID: For implementing DeepSORT algorithm which involves state prediction by utilizing Kalman Filters, data association through Hungarian Algorithm, and deep appearance model of 128 dimensions.
• Optimization of Performance: For optimizing performance without the use of GPUs by skipping frames, Non-Maximum Suppression (NMS), and parameter optimization.

1.2 Scop of the Report

In this report, the entire process related to the development of the internship project is covered, which involves background study, system architecture, mathematics behind YOLOv8 and DeepSORT algorithms, implementation issues, benchmarking and difficulties encountered along with their solutions.

1.3 Company Profile

ALGOBRAIN AI is a growing company providing Artificial Intelligence and Software Solutions and it is located in Surat, Gujarat. Located in 321, Exult Shoppers, Vesu Main Road, close to Siddhi Vinayak Temple, Vesu, Surat (www.algobrainai.com), ALGOBRAIN AI specializes in the areas of AI Chatbots, NLP, Predictive Analytics, Demand Forecasting, and Enterprise Computer Vision.

1.4 Intership Structure and TImeline

• Weeks 1-2 (Orientation & Skill Acquisition): Development environment setup (Python 3.10, PyTorch, OpenCV); research on neural network chatbots and Flask UI application development.

• Weeks 3-4 (Requirement & Detection Model Design): Study on different YOLO models (YOLOv3, YOLOv4, YOLOv8); design of YOLOv8n one stage object detector; video processing workflow.

• Weeks 5-6 (Implementation of DeepSORT & Motion Estimation): Implementation of DeepSORT tracker; setup for 128d Re-ID feature extractor; Kalman Filter state prediction matrices.

• Weeks 7-8 (Optimization, Benchmarking & Documentation): CPU Performance Optimization (Frame Skipping, Non-Maximum Suppression Thresholding); video benchmarking and reporting.

Chapter 2: Literature Review & Problem Statement

Domain Overview

The field of computer vision has seen an exponential increase in development during the last decade, with object detection and tracking being one of the most successful areas of the same. Object detection and tracking in video sequences has led to the development of several revolutionary uses of the same in various fields.

State-of-the-art Review

Early tracking methods were based on simple motion cues using algorithms such as SORT (Simple Online and Realtime Tracking) that utilized Kalman filtering for predicting the state while using the Hungarian algorithm for detection to track association. Although computationally cheap, SORT was highly compromised in crowded scenarios due to the lack of appearance-based re-identification. The DeepSORT, proposed by Wojke et al. (2017), enhanced SORT by introducing a deep appearance descriptor; i.e., CNN pre-trained on large person Re-ID datasets, thus ensuring identity persistence even during occlusion. On the other hand, the evolution of detection frameworks from R-CNN, SSD to different versions of YOLO has drastically reduced computation time.

The introduction of Ultralytics YOLOv8 in 2023 has taken the concept of single-stage detection one step further, offering different model sizes ranging from nano to extra-large models that can be applied across different deployment scenarios.

Technology overview

Technology	Role	Version Used
Python	Core Programming Language	3.10
YOLOv8 (Ultralytics)	Object Detection Model	YOLOv8n
DeepSORT	Multi-Object Tracker	deep_sort_realtime
OpenCV	Video I/O and Visualization	4.x
PyTorch	Deep Learning Backend	2.x
NumPy	Numerical Computing	Latest

Problem Statement

Monitoring of video streams by human operators is very laborious and error-prone. Monitoring of several camera feeds by human operators at the same time leaves them open to the problem of attention fatigue, perceptual errors, and delay responses, thus making real-time monitoring virtually impossible. In the industry, there is a great need for automated solutions capable of detecting, recognizing, and tracking multiple objects through following frames despite various difficulties such as occlusions, disappearances, and reappearance of an object in the camera frame.

Modern solutions of the aforementioned problem require advanced GPUs; however, such systems remain quite expensive for small and medium-sized projects. Keeping the same object in sight from one frame to another becomes a difficult technical task in case of occlusion or collision between several objects. This research project answers the challenges stated above by creating a modular and scalable real-time object tracking pipeline using YOLOv8 and DeepSORT on CPU.

Chapter 3: Methodology

3.1 System Architecture

The current project uses a multi-stage sequential pipeline that can process the video stream in real-time and produce well-labeled output. The language used to implement the project is Python 3.10 due to the variety of its scientific computing and computer vision libraries. Regarding video input/output, frame capturing and results visualization like bounding boxes and labels for objects tracking, the library OpenCV is used in the current project. As for deep learning computations, the library PyTorch which makes tensor calculations and runs models on standard CPU is the main one used in the current project.

Regarding the pipeline structure, the following steps are implemented: Video Input (WebCam or MP4) → Frame Capturing → Object Detection (YOLOv8) → Feature Extraction (Re-ID embeddings) → Multi-object Tracking (DeepSORT using Kalman filter and Hungarian algorithm) → Visualization Output.

3.2 Object Detection Phase (YOLOv8)

The UltraLycatrics YOLOv8n (nano) model has been used as the core object detection system. Specifically, the nano model is preferred for its speed and weight characteristics because it is the fastest and smallest YOLOv8 model that can be run in real-time on a typical CPU. It should be noted that YOLOv8 works with the one-pass Convolutional Neural Network (CNN), which means that the entire input image is processed in a single pass through the network instead of using the two-stage detectors with their region proposal process.

YOLOv8 divides the input image into a grid of cells, each of which predicts bounding boxes of detected objects in the format (x, y, w, h) – where x and y represent the coordinates of the center point of the detected object and w, h – width and height, respectively. Also, the bounding box is supplemented by confidence scores and class probability scores. CSPDarknet architecture is responsible for feature extraction, while the PANet architecture is used for multi-scale feature map aggregation to detect objects of different sizes.

3.3 Feature Extraction and Traking (DeepShort)

The DeepSORT improves upon the SORT method by introducing deep appearance features using the Re-ID Convolutional Neural Network. Appearance feature vector obtained is of 128 dimensions. The position of the bounding box for each of the objects detected is estimated in the future using Kalman filter, whereas the Hungarian algorithm is applied to match predictions with the newly detected object using cosine distance.

There are three types of Track states that include:

- Tentative
- Confirmed
- Deleted

The track state changes from Tentative to Confirmed when it stays consistent for sometime, and changes to Deleted when the track state goes without detection more than max_age.

3.4 Software Optimization

To combine both the algorithms, the while loop constantly fetches frames from OpenCV. Boxes which have their confidence score higher than 0.5 are passed to the Deep SORT tracker. In the case of max_age, 30 was chosen as its value, meaning that a certain track will survive for 30 frames after which there is no detection, after which it dies permanently. Frame skipping was used for real-time performance.

Challenge	Solution Implemented
ID Switching on Occlusion	DeepSORT cascade matching + Re-ID CNN appearance features
Low FPS on CPU	Switched to YOLOv8n nano; applied frame skipping; achieved 25+ FPS
False Detections	Confidence threshold (conf > 0.5); NMS with IoU threshold of 0.45
Track Loss in Fast Motion	Tuned Kalman filter noise; max_age=30; input resolution 640×640

Chapter 4: Results and Discussion

4.1 Detection and Perfomance Matrix

The system was tested on multiple video datasets in real-life scenarios. The system was capable of achieving 89% mAP for object detection. This system was capable of sustaining an output speed of 25-30 FPS utilizing the YOLOv8-nano algorithm along with frame skipping technique, even on the regular CPU.

Metric	Value	Remarks
Detection Accuracy (mAP)	89%	Evaluated on real-world video benchmarks
Processing Speed (FPS)	25–30 FPS	Standard CPU, real-time performance
Track ID Consistency	95%	Stable IDs maintained across frames
Avg. Processing Latency	<40 ms/frame	Including detection and tracking overhead

4.2 Tracking Reliability

The key performance indicators used in this project were viewed to be the stability and identity consistency since the main issue which is solved by DeepSORT algorithm is the issue of keeping identity consistent across consecutive frames. DeepSORT tracker reached 95% Track ID Consistency for all tested videos which is rather high result compared to SORT trackers which are very prone to switching identities in case of objects occlusion.

DeepSORT tracker managed to successfully track the whole track life cycle starting from Tentative state to Confirmed state and then to Deletion so that all objects which appeared after being occluded by any obstacle or something else got the same ID number instead of being considered as new objects. This was especially noticeable when several pedestrians were moving at the same time and 128D Re-ID appearance vectors worked as the key discriminative information needed for assigning detection to the right track. The max_age parameter equal to 30 frames was also helpful in this case since it allowed more time to recover lost tracks.

4.3 Overview Technical challenges

Switching IDs in case there was an overlap of objects during training was a major challenge. DeepSORT's Re-ID CNN was used to resolve this challenge by using visual feature matching with 128D. In order to resolve the challenge of false detections, NMS with IoU value 0.45 was utilized.

4.4 Key Takeaways

The success in the deployment of this system has led to some valuable lessons regarding both technical and operational aspects of the approach that go far beyond the context of the created solution itself. The results have conclusively shown that the use of a light-weight detection system, such as YOLOv8n, in tandem with an advanced appearance-based tracking mechanism, such as DeepSORT, allows one to achieve true real-time multi-object tracking on regular CPU equipment, thus making intelligent video analysis widely accessible to everyone, eliminating the need for GPU-based computation.

The essential role of appearance-related features in tracking tasks has been firmly validated by the present project. Indeed, whenever the targets being tracked often intersect each other, get obscured, or even temporarily leave the camera's field of view, any tracking method that relies solely on their motion is doomed to fail. The 128-dimension appearance feature vectors generated by DeepSORT's CNN-based appearance model have proven themselves indispensable in the cases mentioned above. Moreover, the fine-tuning of the various system parameters, namely the Kalman filter process noise, max_age threshold, confidence filtering value, and input resolution, have proven to be crucial for the quality of the tracking result achieved.

4.5 Future Work

Hardware Installation:
• Install all the components in edge computing hardware such as Raspberry Pi and NVIDIA Jetson Nano.
Model Fine-Tuning:
• Fine-tune the YOLOv8 model with custom data based on the application context (e.g., dense crowds and highways).
Heatmaps:
• Include heatmap visualization for object movements, dwell times, and traffic bottlenecks.
Extended Features:
• Extend the system with additional features such as facial recognition and stitching camera images into one image.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

The project managed to create a very efficient multi-object tracking application with the help of the seamless integration of the object detection power of YOLOv8 with the DeepSORT multi-object tracker. The created application managed to overcome the common limitations of surveillance systems like errors, changing of the IDs in case of object occlusions, and high cost due to the computational load. The application proved itself to be very efficient with very good metrics like 89% Mean Average Precision (mAP), 95% Track ID Consistency, and consistent 25-30 FPS on regular CPU without GPU. The results obtained provide a clear confirmation of the feasibility of the real-time computer vision on commodity hardware.

Thanks to the modular structure of the pipeline used, it is possible to achieve a great scalability of the application, which means that each part of the system can be replaced or updated without affecting the other elements. It makes the created project the foundation of many practical applications in the spheres of automated surveillance, traffic control, retail analytics, and crowd monitoring systems. In conclusion, the internship project allowed not only improving skills of working with deep learning models and computer vision but also creating a real-life system.

5.2 Recommendations

Taking into consideration the findings and problems encountered in this process, the following recommendations can be provided for the further improvement of the technology. First of all, concerning the hardware part, it is recommended that the pipeline be used in edge computing devices such as Raspberry Pi 5 or NVIDIA Jetson Nano to ensure low-latency processing without cloud service involvement. As far as model fine tuning goes, YOLOv8 can be fine-tuned using custom datasets, including dense crowd movement, highway traffic, or retail shelf monitoring to boost its performance in these areas.

As far as analytics go, it would be helpful to include the functionality of generating heatmaps of the object's previous locations, time spent there, and traffic hot spots to turn the tracking system into a proactive decision-making tool rather than reactive. In addition to that, the addition of facial recognition or full body Re-ID models would help increase the accuracy of person tracking if there were changes in people's appearances while moving from camera to camera. Lastly, it is recommended to scale up the solution with multi-camera stitching and REST API layer to enable cloud dashboard live streams.

APPENDICES

Appendix A : Letter of comfirmation--Internship

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

INTERNSHIP OFFER LETTER

Dear Dhruv Hiteshbhai Trivedi ,

We are pleased to offer you a 2 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location : Hybrid / India
- Duration : 01/06/2026 to 01/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/06/2026

Dr. Anurag Shrivastava
Talent Acquisition Manager

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Appendix B : InternshipCompletion Letter

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NATIONAL INTERNSHIP PORTAL

MINISTRY OF EDUCATION

CERTIFICATE

OF INTERNSHIP

This certificate is awarded to

Dhruv Hiteshbhai Trivedi

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Dr. Anurag Shrivastava

Program Manager